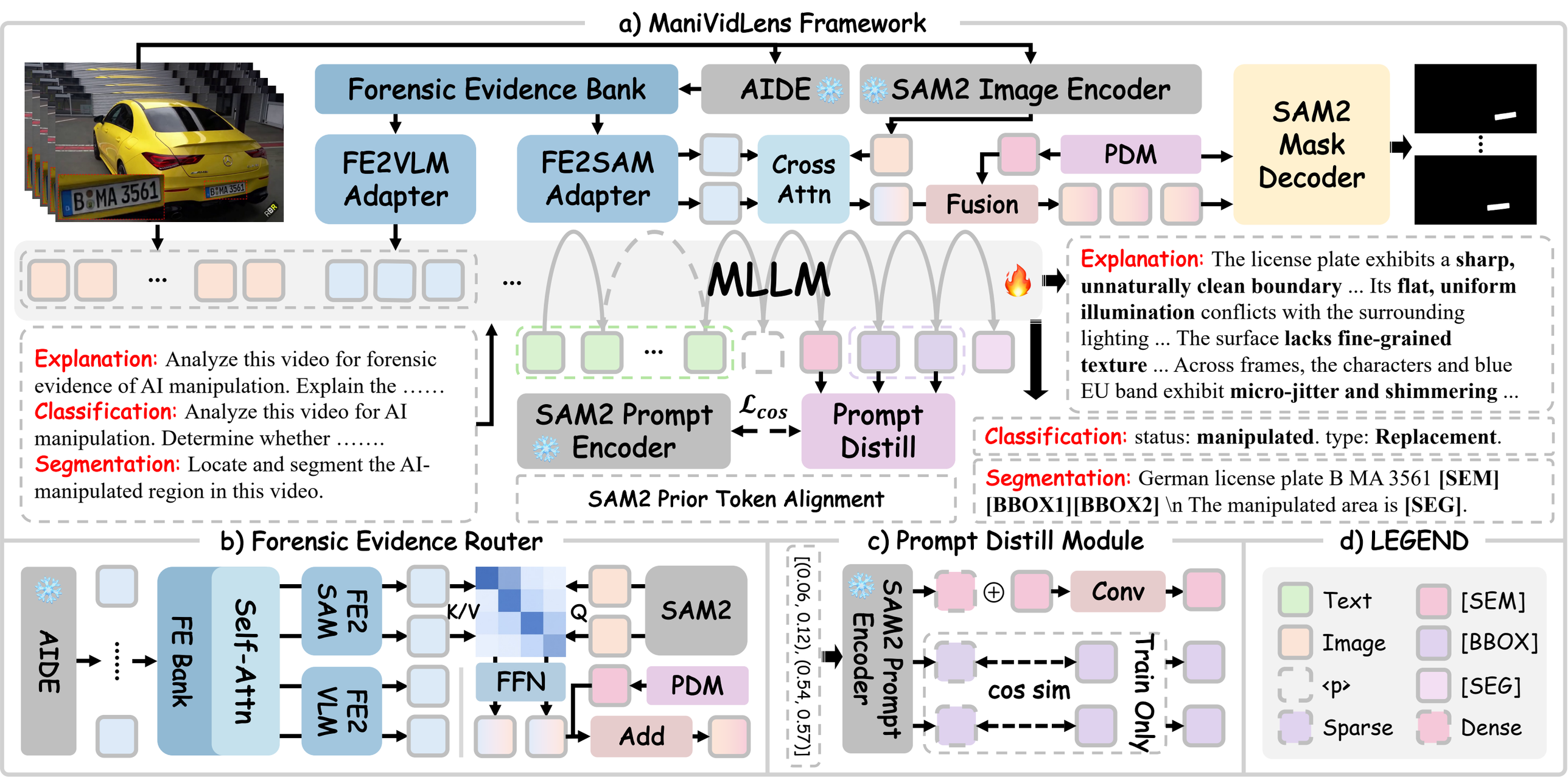

a) ManiVidLens Framework
Forensic Evidence Bank
AIDE
SAM2 Image Encoder
FE2VLM Adapter
FE2SAM Adapter
Cross Attn
PDM
Fusion
SAM2 Mask Decoder
MLLM
Explanation: Analyze this video for forensic evidence of AI manipulation. Explain the ……
Classification: Analyze this video for AI manipulation. Determine whether …….
Segmentation: Locate and segment the AI-manipulated region in this video.
SAM2 Prompt Encoder
$\mathcal{L}_{cos}$
Prompt Distill
SAM2 Prior Token Alignment
Explanation: The license plate exhibits a sharp, unnaturally clean boundary ... Its flat, uniform illumination conflicts with the surrounding lighting ... The surface lacks fine-grained texture ... Across frames, the characters and blue EU band exhibit micro-jitter and shimmering ...
Classification: status: manipulated. type: Replacement.
Segmentation: German license plate B MA 3561 [SEM] [BBOX1][BBOX2] \n The manipulated area is [SEG].
b) Forensic Evidence Router
AIDE
FE Bank
Self-Attn
FE2 SAM
FE2 VLM
K/V
Q
FFN
SAM2
PDM
Add
c) Prompt Distill Module
[(0.06, 0.12), (0.54, 0.57)]
SAM2 Prompt Encoder
Conv
cos sim
Train Only
d) LEGEND
Text
Image
<p>
Sparse
[SEM]
[BBOX]
[SEG]
Dense